Multivariable calculus · Differentiation in two variables

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Partial derivatives and directional slopes

A partial derivative changes one coordinate while holding the others fixed. It is the slope of the corresponding surface section. The gradient collects these coordinate slopes, and its dot product with a unit vector gives the directional derivative. Mixed partials agree under suitable continuity assumptions.

S1

2 · Total change and the chain rule

For a differentiable function, the total differential combines the coordinate contributions to first-order change. Along a differentiable path, the chain rule gives the exact derivative of the composition. Check the result by direct substitution when that route is practical.

S2

3 · Implicit curves and valid tangent formulas

Implicit differentiation applies the chain rule to a relation. For a regular branch with nonzero F_y, the slope is minus F_x divided by F_y. The equation itself validates the evaluation point. A zero denominator may indicate that x should be treated as a function of y instead.

S3

2 Keyword index

9 terms; the number in brackets is the topic.

Chain rule (2). Derivative of a composition through all changing inputs. S2

Differential (2). Linear change at a specified base point. S2

Directional derivative (1). Rate along a specified unit direction. S1

Gradient (1). Vector of coordinate partial derivatives. S1

Implicit function (3). Local function represented by a relation. S3

Mixed partial (1). Derivative taken with respect to different coordinates. S1

Regular point (3). Here, a point where the gradient of the relation is nonzero. S3

Remainder (2). Difference between an exact increment and a chosen approximation. S2

Vertical tangent (3). Tangent parallel to the y-axis. S3

3 Rapid recall

1	Why can the same surface have two coordinate slopes?	An x-section and a y-section follow different curves through the same point, so their slopes need not agree. S1
2	Why does a differential differ from a finite change?	The differential keeps the linear part. Nonlinear terms remain in the exact finite increment. S2
3	Why does a zero denominator require a new analysis?	It prevents division in the chosen formula. The curve may instead have a vertical tangent or fail to be regular; examine the defining equation. S3

4 Last 10 minutes before the exam

Top facts

1. Differentiate before substituting the evaluation point. Substitution too early can erase the variable that must change. S1
2. A path can change several inputs at once; include one chain-rule term for each changing input. S2
3. Verify the point belongs to the relation and the chosen denominator is nonzero before evaluating an implicit derivative. S3

Top formulas

E1.1 · Coordinate derivatives

$$f_x = 2xy, \qquad f_y = x^2 + \cos y$$

GIVES	Coordinate slopes for $f = x^2y + \sin y$.
APPLIES	Hold the other independent coordinate fixed.
UNITS	Output/input; dimensionless in this example.

S1

E2.2 · Chain rule

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

GIVES Change along a parameterised path.

APPLIES Differentiate every changing input. Evaluate partials on the path.

UNITS Output per parameter unit.

S2

E3.1 · Implicit first derivative

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

GIVES Slope of a regular implicit branch.

APPLIES $F = 0$ at the point and $F_y \neq 0$.

UNITS Units of y divided by units of x.

S3